

Airframe PID Parameter Comparison

All 12 airframes from `originaldefaults.h` converted to unified float `.af` format. Values are in FC-native units (radians, volts, fractions — not degrees, volts×10, or percent).

Rate PID Gains

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450- 1165	Ken's Alpha	Ken's LadyBug	Phoenix FW	Rok's Quad	S500- 1137	Sky Surfer	Spoileron	Wing 900mm
RollRateKp	0.16	0.095	0.1	0.16	0.095	0.16	0.04	0.095	0.1	0.06	0.2	0.04
RollRateKd	0.006	0.004	0.0045	0.006	0.004	0.006	0	0.004	0.0045	0	0.004	0

Angle PID Gains

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450- 1165	Ken's Alpha	Ken's LadyBug	Phoenix FW	Rok's Quad	S500- 1137	Sky Surfer	Spoileron	Wing 900mm
RollAngleKp	7	5.75	7	7	5.75	7	2	5.75	7	3	0.75	2
RollAngleKi	0.1	0.15	0.25	0.2	0.15	0.1	0.1	0.15	0.25	0.15	0.1	0.1
RollAngleIntLimit	0.005236	0.002618	0.002618	0.00733	0.002618	0.005236	0.002618	0.002618	0.002618	0.002618	0.002618	0.002618

Pitch Axis

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450- 1165	Ken's Alpha	Ken's LadyBug	Phoenix FW	Rok's Quad	S500- 1137	Sky Surfer	Spoileron	Wing 900mm
PitchRateKp	0.16	0.095	0.1	0.16	0.095	0.16	0.04	0.095	0.1	0.1	0.2	0.04
PitchRateKd	0.006	0.004	0.0045	0.006	0.004	0.006	0	0.004	0.0045	0	0.004	0
PitchAngleKp	7	5.75	7	7	5.75	7	2	5.75	7	7.5	0.75	2
PitchAngleKi	0.1	0.15	0.25	0.1	0.15	0.1	0.1	0.15	0.25	0.4	0.1	0.1
PitchAngleIntLimit	0.005236	0.002618	0.002618	0.00733	0.002618	0.005236	0.002618	0.002618	0.002618	0.003927	0.002618	0.002618

Yaw Axis

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450- 1165	Ken's Alpha	Ken's LadyBug	Phoenix FW	Rok's Quad	S500- 1137	Sky Surfer	Spoileron	Wing 900mm
YawRateKp	0.2	0.125	0.1	0.2	0.125	0.2	0.05	0.125	0.1	0.05	0.05	0.05
YawRateKd	0.000125	0.001125	0.001125	0.001125	0.001125	0.000125	0.001125	0.001125	0.001125	0.001125	0.001125	0.001125
YawAngleKp	2.25	3	3	3	3	2.25	3	3	3	3	3	3
YawAngleKi	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
YawAngleIntLimit	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727

Navigation Gains

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450- 1165	Ken's Alpha	Ken's LadyBug	Phoenix FW	Rok's Quad	S500- 1137	Sky Surfer	Spoileron	Wing 900mm
NavPosKp	0.33	0.2475	0.2475	0.2475	0.2475	0.33	0.33	0.2475	0.33	0.33	0.33	0.33
NavPosKi	0.008	0.012	0.012	0.012	0.012	0.008	0.008	0.012	0.008	0.008	0.008	0.008
NavVelKp	0.6	0.6	0.3	0.6	0.6	0.6	0.6	0.3	0.6	0.6	0.6	0.6

Altitude Gains

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450- 1165	Ken's Alpha	Ken's LadyBug	Phoenix FW	Rok's Quad	S500- 1137	Sky Surfer	Spoileron	Wing 900mm
AltPosKp	0.5124	0.5124	0.5124	0.1464	0.4575	0.915	0.2928	0.4392	0.5124	0.2928	0.2928	0.2928
AltPosKi	0.0046	0.0046	0.0046	0.00092	0.00184	0.0138	0.00552	0.00276	0.0046	0.00552	0.00552	0.00552

Cruise & Limits

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450- 1165	Ken's Alpha	Ken's LadyBug	Phoenix FW	Rok's Quad	S500- 1137	Sky Surfer	Spoileron	Wing 900mm
EstCruiseThr	0.59	0.3	0.54	0.31	0.3	0.54	0.2	0.5	0.55	0.2	0.2	0.2
MaxRollAngle	1.0472	1.0472	0.523599	1.0472	1.0472	1.0472	1.0472	1.0472	1.0472	1.0472	1.0472	1.0472
MaxPitchAngle	1.0472	1.0472	0.523599	1.0472	1.0472	1.0472	1.0472	1.0472	1.0472	0.523599	1.0472	1.0472
NavMaxAngle	0.261799	0.349066	0.436332	0.349066	0.349066	0.261799	0.349066	0.261799	0.349066	0.698132	0.349066	0.349066
BatteryCapacity	22	22	22	22	22	22	22	50	22	22	22	22
LowVoltThres	3.4	9.6	9.6	13.6	13.8	3	10.2	10	10.2	10.2	10.2	10.2
RCChannels	7	7	7	7	7	7	7	7	7	7	7	7
MaxRollRate	10.472	10.472	10.472	10.472	10.472	10.472	10.472	10.472	10.472	10.472	10.472	10.472
MaxPitchRate	10.472	10.472	10.472	10.472	10.472	10.472	10.472	10.472	10.472	10.472	10.472	10.472